C. Quotient and Remainder

time limit per test 2 seconds
memory limit per test 512 megabytes
input standard input
output standard output

Rating: 1300 | Tags: binary search, greedy, two pointers | https://codeforces.com/problemset/problem/2170/C

You are given two integer arrays: $q_1, q_2, \ldots, q_n$ and $r_1, r_2, \ldots, r_n$, as well as an integer k.

You can perform the following operation any number of times (possibly zero):

1. Choose two integers x and y such that
- $1 \le y$ lt $x \le k$;
- there exists an index i such that $q_i = \lfloor x/y \rfloor$ (rounded down);
- there exists an index j such that $r_j = x \bmod y$.
2. Remove q_i from the array q and r_j from the array r. If there are multiple occurrences of q_i in the array q, only one occurrence is removed; same for r_j and the array r.

Calculate the maximum number of operations that you can perform on the given arrays q and r.

Input

The first line contains one integer t ($1 \le t \le 10^4$) — the number of test cases.

The first line of each test case contains two integers n and k ($1 \le n \le 2 \cdot 10^5$; $2 \le k \le 10^{18}$) — the size of the arrays q and r and the upper limit for x and y.

The second line of each test case contains n integers $q_1, q_2, \ldots, q_n$ ($1 \le q_i \le 10^9$) — the array q.

The third line contains n integers $r_1, r_2, \ldots, r_n$ ($1 \le r_i \le 10^9$) — the array r.

Additional constraints on the input: the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, print one integer — the maximum number of operations that you can perform on the given arrays.

Example

Input
3 1 100 1 27 3 10 5 6 5 7 1 7 5 42 5 4 2 2 1 9 8 9 8 100

Output
1 0 3

Note

In the first test case, one operation can be performed: you can choose $x = 69$ and $y = 42$. Then $\lfloor (69)/(42) \rfloor = 1 = q_1$ and $69 \bmod 42 = 27 = r_1$.

In the second test case, it is impossible to perform any operations, as suitable x and y cannot be found.

In the third test case, three operations can be performed:

1. $x = 42$, $y = 17$: $\lfloor (42)/(17) \rfloor = 2 = q_3$ and $42 \bmod 17 = 8 = r_2$;
2. $x = 41$, $y = 16$: $\lfloor (41)/(16) \rfloor = 2 = q_4$ and $41 \bmod 16 = 9 = r_1$;
3. $x = 20$, $y = 12$: $\lfloor (20)/(12) \rfloor = 1 = q_5$ and $20 \bmod 12 = 8 = r_4$;

After these operations, you'll get arrays $q = [5, 4]$ and $r = [9, 100]$, and there are no more operations you can perform.